

MEDICAL DIAGNOSTIC TEST CENTER PORTAL

A ServiceNow-Based Application for Digitizing Diagnostic Center Operations

A PROJECT REPORT

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[KRCE/CSE(AIML)]

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Medical Diagnostic Test Center Portal

ServiceNow Project Documentation

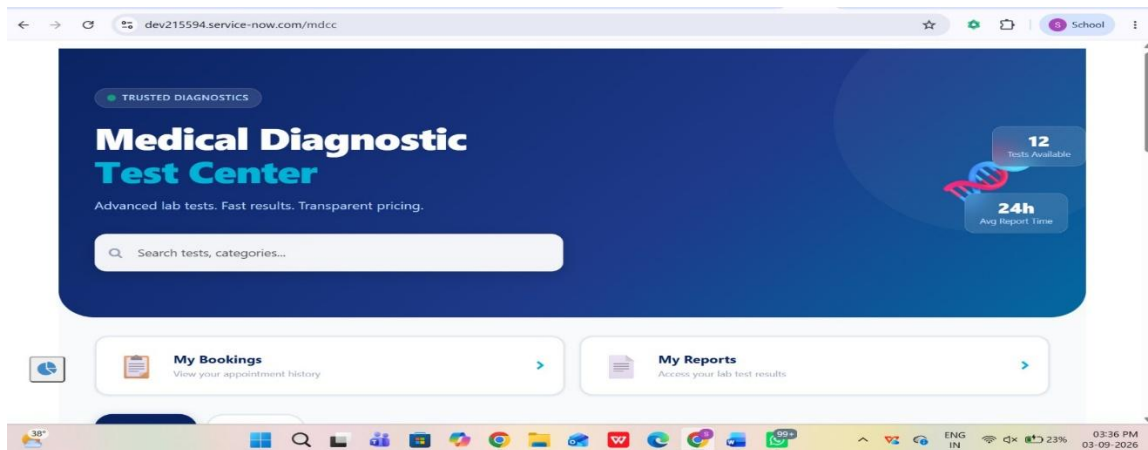
1. Introduction

The **Medical Diagnostic Test Center Portal** is a ServiceNow-based application designed to digitize and automate the diagnostic test appointment management process.

The system provides a centralized platform where patients can browse diagnostic tests, check appointment availability, book appointments, make payments, track appointment status, and securely access diagnostic reports.

The system also supports receptionists, laboratory technicians, and administrators by automating appointment processing, report management, notifications, and access control.

The application is developed using ServiceNow features such as **Service Portal, Custom Tables, Client Scripts, Business Rules, Flow Designer, Notifications, ACLs, Reports, and Dashboards.**



2. Problem Statement

Traditional diagnostic center operations often involve manual appointment booking, phone-based communication, paper-based reports, and limited appointment tracking.

The major problems include:

- Manual appointment scheduling
- Difficulty checking available time slots
- Duplicate appointment bookings
- Delays in report delivery
- Lack of centralized appointment tracking
- Manual notification processes

- Unauthorized access to medical information
- Difficulty managing cancellations and refunds

3. Objectives

The main objectives of the system are:

1. To provide online diagnostic test booking.
2. To display available appointment slots dynamically.
3. To prevent duplicate appointment bookings.
4. To automate appointment processing using Flow Designer.
5. To provide secure diagnostic report management.
6. To automate email notifications.
7. To implement role-based access control.
8. To provide appointment status tracking.
9. To reduce manual administrative work.
10. To provide a scalable platform for diagnostic center operations.



4. System Users

The system supports four major user roles.

User	Responsibilities
Patient	Select tests, book appointments, make payments, view appointments and reports
Receptionist	Manage appointments, approvals, cancellations and patient requests
Lab Technician	View assigned appointments, update sample status and upload reports
Administrator	Manage system configuration, users, security, tables, workflows and reports

5. Technology Stack

Component	Technology
Platform	ServiceNow
Frontend	Service Portal
UI	HTML, CSS, AngularJS
Backend	ServiceNow Server-side scripting
Database	ServiceNow Tables
Automation	Flow Designer
Business Logic	Business Rules / Scripts
Client-side Validation	Client Scripts
Security	ACLs and Roles
Notifications	ServiceNow Email Notifications
Reporting	Reports & Dashboards
Deployment	Update Sets

6. Prerequisites

Before implementing the application, the following requirements must be available.

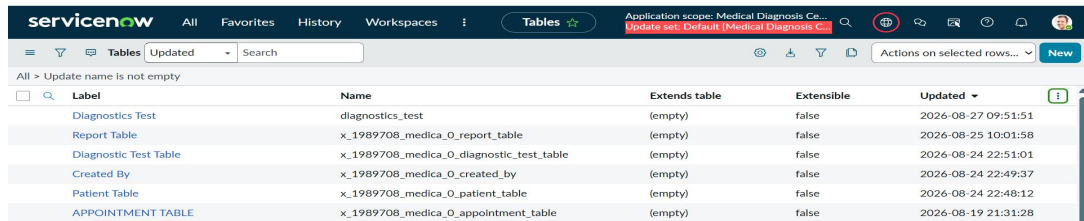
ServiceNow Requirements

- Active ServiceNow instance
- Service Portal enabled
- Studio / Scoped Application Development access
- Administrator access
- Flow Designer access
- ACL configuration access
- Business Rule configuration access
- Email Notification access
- Reports and Dashboards access

Required Roles

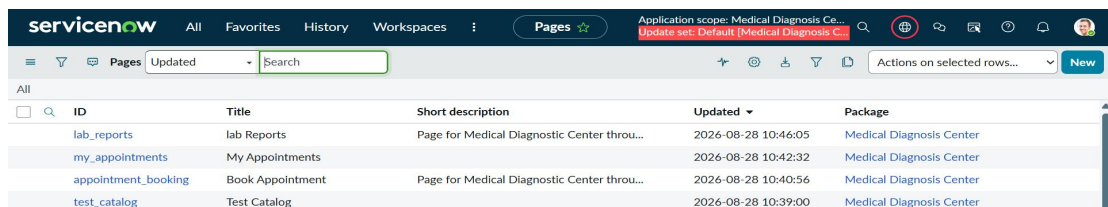
The administrator should have the required permissions to create and configure:

- Tables



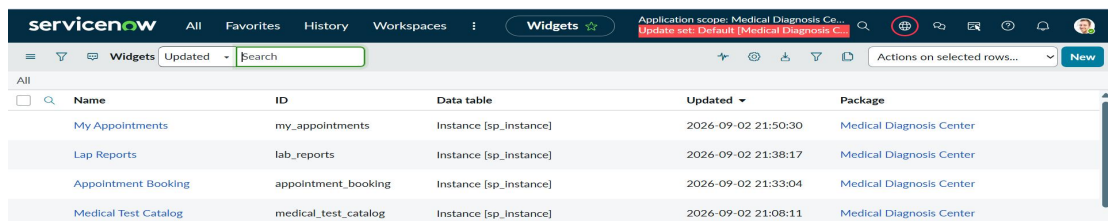
Label	Name	Extends table	Extensible	Updated
Diagnostics Test	diagnostics_test	(empty)	false	2026-08-27 09:51:51
Report Table	x_1989708_medica_0_report_table	(empty)	false	2026-08-25 10:01:58
Diagnostic Test Table	x_1989708_medica_0_diagnostic_test_table	(empty)	false	2026-08-24 22:51:01
Created By	x_1989708_medica_0_created_by	(empty)	false	2026-08-24 22:49:37
Patient Table	x_1989708_medica_0_patient_table	(empty)	false	2026-08-24 22:48:12
APPOINTMENT TABLE	x_1989708_medica_0_appointment_table	(empty)	false	2026-08-19 21:31:28

- Fields
- Forms
- Service Portal pages



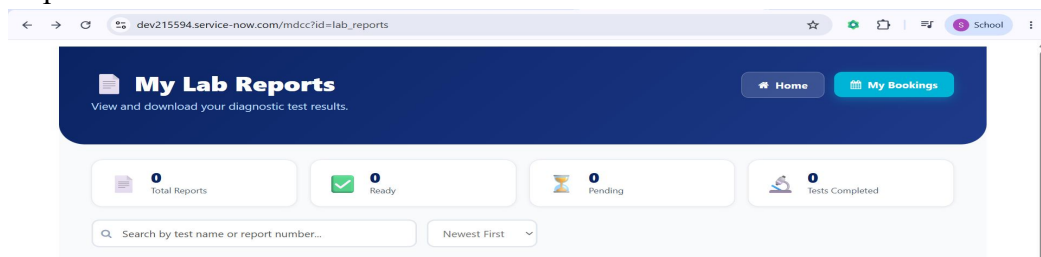
ID	Title	Short description	Updated	Package
lab_reports	lab Reports	Page for Medical Diagnostic Center throu...	2026-08-28 10:46:05	Medical Diagnosis Center
my_appointments	My Appointments		2026-08-28 10:42:32	Medical Diagnosis Center
appointment_booking	Book Appointment	Page for Medical Diagnostic Center throu...	2026-08-28 10:40:56	Medical Diagnosis Center
test_catalog	Test Catalog		2026-08-28 10:39:00	Medical Diagnosis Center

- Widgets



Name	ID	Data table	Updated	Package
My Appointments	my_appointments	Instance [sp_instance]	2026-09-02 21:50:30	Medical Diagnosis Center
Lap Reports	lab_reports	Instance [sp_instance]	2026-09-02 21:38:17	Medical Diagnosis Center
Appointment Booking	appointment_booking	Instance [sp_instance]	2026-09-02 21:33:04	Medical Diagnosis Center
Medical Test Catalog	medical_test_catalog	Instance [sp_instance]	2026-09-02 21:08:11	Medical Diagnosis Center

- Client Scripts
- Business Rules
- Flows
- Notifications
- ACLs
- Reports



7. System Architecture

The application follows a layered architecture.

User Layer

Patients, Receptionists, Lab Technicians and Administrators interact with the application.



Service Portal Layer

The Service Portal provides:

- Homepage
- Test catalog
- Appointment booking
- Appointment tracking
- Report access
- Payment interface



Application Layer

ServiceNow application logic handles:

- Validation
- Pricing
- Slot management
- Appointment processing
- Notifications
- Report processing



Automation Layer

Flow Designer and Business Rules automate:

- Appointment processing
- Approvals
- Payment confirmation
- Status updates

- Notifications
- Cancellation processing



Data Layer

Custom ServiceNow tables store:

- Patient information
- Diagnostic tests
- Appointments
- Reports
- Payments

8. Custom Table Configuration

The application uses multiple custom tables to maintain structured information.

8.1 Patient Table

The Patient table stores patient information.

Important fields

- Patient ID
- Patient Name
- Date of Birth
- Gender
- Contact Number
- Email
- Address

A unique auto-generated Patient ID can be configured using the prefix:

PAT

8.2 Diagnostic Test Catalog

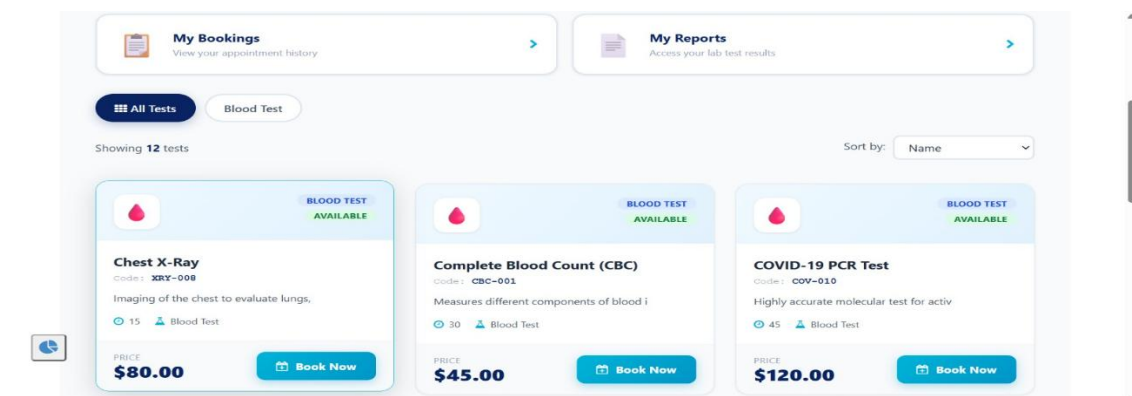
This table stores the diagnostic tests offered by the center.

Fields

- Test ID
- Test Name
- Test Description
- Test Category
- Price
- Duration
- Availability
- Status

Example:

Test	Price	Status
Blood Test	₹500	Active
X-Ray	₹800	Active
MRI	₹3000	Active



8.3 Appointment Details

This table manages patient appointments.

Fields

- Appointment Number
- Patient
- Diagnostic Test
- Appointment Date
- Appointment Time
- Status
- Price
- Payment Status
- Assigned Technician
- Created Date

Appointment numbers can be automatically generated using:

APPT



The screenshot shows a web application interface for the 'APPOINTMENT TABLE'. At the top, there is a header bar with a back arrow, a menu icon, the title 'APPOINTMENT TABLE', a timestamp 'Created 2026-09-02 21:44:15', and action buttons for 'Update' and 'Delete'. Below the header, the form contains several input fields: 'Appointment Number' with the value 'APPT-2026-51345', 'Approval Date' with a calendar icon, 'Status' with a dropdown menu showing 'pending', and 'Approval Status' with a dropdown menu showing 'requested'.

8.4 Report Records

The Report table manages diagnostic reports.

Fields

- Report Number
- Appointment
- Patient
- Test
- Report File

- Uploaded By
- Upload Date
- Report Status

Report numbers can use the prefix:

REP

8.5 Payment Table

The Payment table stores payment-related information.

Fields

- Payment ID
- Appointment
- Patient
- Amount
- Payment Status
- Payment Date
- Transaction ID


Payment

Select a payment method and confirm your booking.

Order Summary

ECG (Electrocardiogram)	\$55.00
Date & Time	Tuesday, 8 Sep 2026 - 01:00 PM
Service Fee	\$0.00
Total	\$55.00

SELECT PAYMENT METHOD

 Card

 UPI

 Net Banking

 Insurance

 Pay on Arrival



Pay on Arrival

Pay **\$55.00** at the center when you arrive.

Carry a valid ID. Please arrive 15 minutes early.

← Back

Confirm & Pay \$55.00

9. Reference Relationships

Relationships between tables are established using reference fields.

Patient



Appointment

- Diagnostic Test
- Payment
- Report

10. Service Portal Configuration

A dedicated Service Portal is created for the Diagnostic Test Center.

Portal Configuration

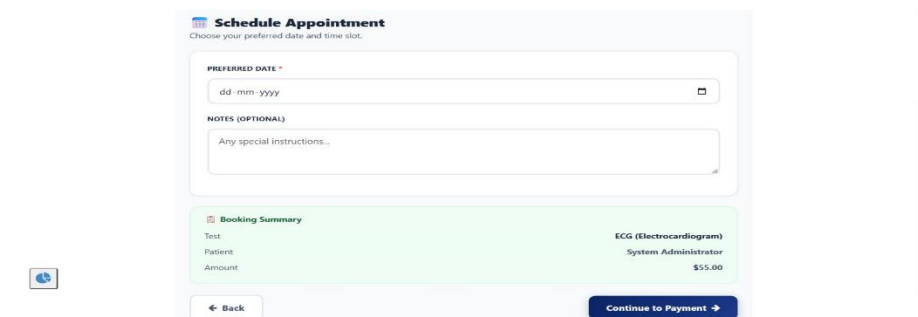
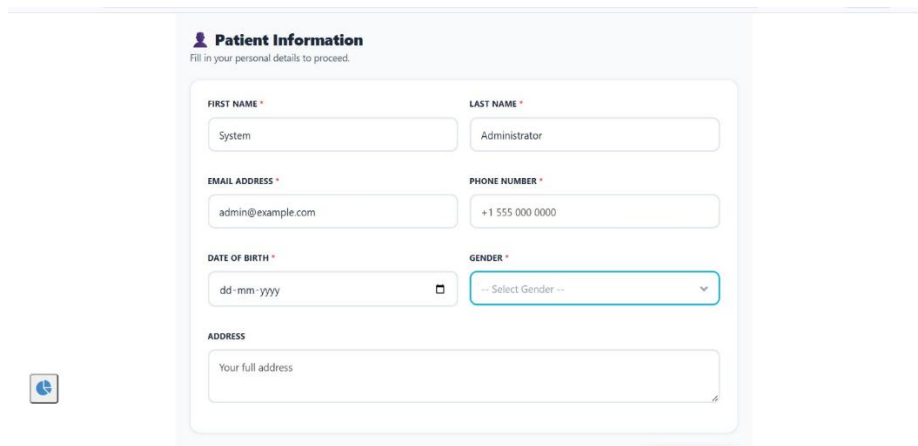
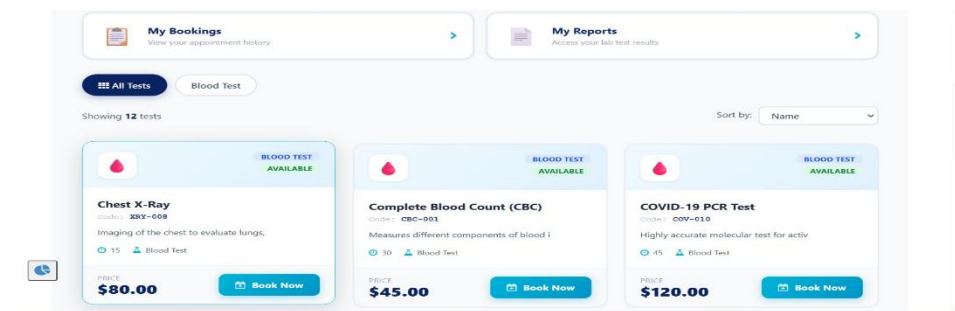
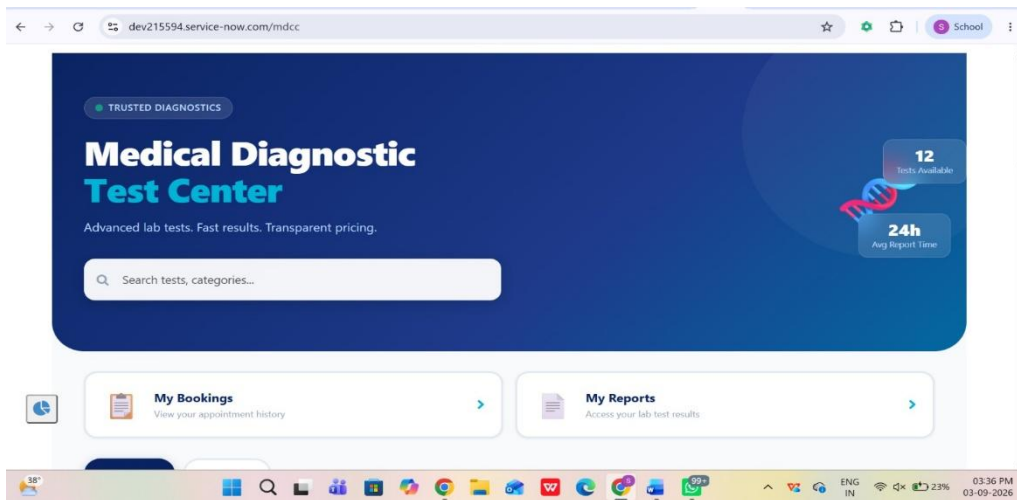
The following components are configured:

- Portal title
- Logo
- Theme
- Homepage
- Navigation menu
- Footer
- Search functionality

Custom Pages

The portal contains pages such as:

1. Test Selection
2. Appointment Booking
3. My Appointments
4. Report Download
5. Payment
6. Booking Confirmation



[← Back](#)

[View My Bookings](#)

Book Another Test

11. Custom Widget Development

Custom Service Portal widgets are developed to provide an interactive user interface.

Technologies Used

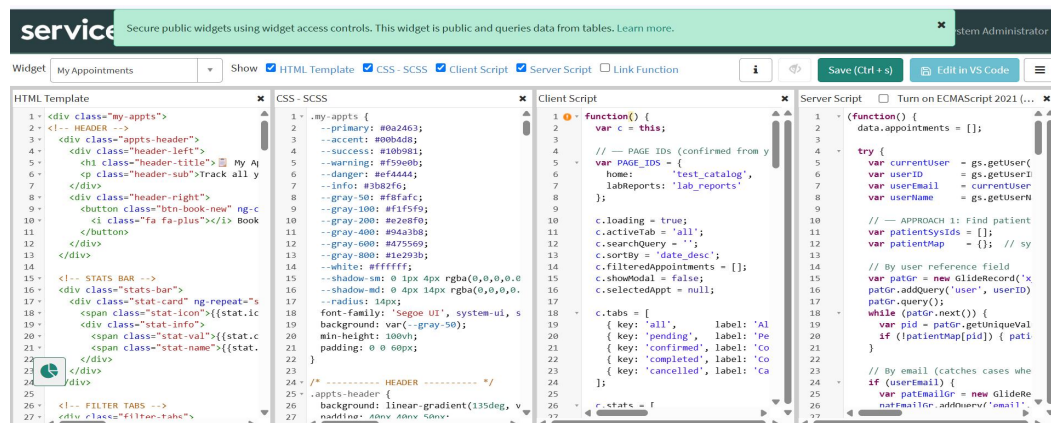
- HTML
- CSS
- AngularJS
- Server Script
- Client Controller

Major Widget Functions

The widgets provide:

- Diagnostic test display
- Dynamic slot availability
- Appointment booking
- Price calculation
- Booking confirmation
- Appointment tracking
- Report visibility
- Report download

The widgets are tested using **Service Portal Preview** before deployment.



12. Appointment Booking Process

The appointment booking process follows these steps:

Patient Login



Select Diagnostic Test



Select Date



Display Available Slots



Select Time Slot



Validate Appointment



Calculate Price



Create Appointment



Payment



Confirmation

The screenshot shows a web form titled "Schedule Appointment" with the subtitle "Choose your preferred date and time slot:". The form contains a "PREFERRED DATE" field with a date picker icon, a "NOTES (OPTIONAL)" text area with placeholder text "Any special instructions...", and a "Booking Summary" section. The summary table lists the test as "ECG (Electrocardiogram)", the patient as "System Administrator", and the amount as "\$55.00". At the bottom, there are "Back" and "Continue to Payment" buttons.

Booking Summary	
Test	ECG (Electrocardiogram)
Patient	System Administrator
Amount	\$55.00

Schedule Appointment
Choose your preferred date and time slot.

PREFERRED DATE: 09-09-2026

AVAILABLE TIME SLOTS:

08:00 AM	08:30 AM	09:00 AM	09:30 AM	10:00 AM	10:30 AM	11:00 AM
11:30 AM	12:00 PM	12:30 PM	01:00 PM	01:30 PM	02:00 PM	02:30 PM
03:00 PM	03:30 PM	04:00 PM	04:30 PM	05:00 PM		

NOTES (OPTIONAL)
Any special instructions...

Booking Summary

Test	ECG (Electrocardiogram)
Patient	System Administrator
Date	Monday, 7 Sep 2026
Time	01:00 PM
Amount	\$55.00

Back Continue to Payment

Validation

The system verifies:

- Mandatory fields
- Valid appointment date
- Available slot
- Duplicate booking
- Test availability
- Patient information

If validation fails, the user receives an appropriate error message.

13. GlideRecord Implementation

GlideRecord is used for server-side database operations.

It can be used to:

- Retrieve patient records
- Check existing appointments
- Create new appointments
- Update appointment status
- Retrieve reports
- Validate slot availability

For example, before creating an appointment, the system can query existing appointment records to determine whether the selected date and time slot is already occupied.

14. Flow Designer Automation

Flow Designer is used to automate the application workflow.

Main Flows

Flow 1 – Appointment Booking

Trigger:

New appointment created.

Actions:

- Validate appointment
- Process booking
- Update status
- Send confirmation notification

Flow 2 – Approval

Trigger:

Appointment requires approval.

Actions:

- Send approval request
- Wait for approval
- Update appointment status
- Notify patient

Flow 3 – Report Availability

Trigger:

Diagnostic report uploaded.

Actions:

- Update report status
- Update appointment status
- Send notification to patient

Flow 4 – Cancellation and Refund

Trigger:

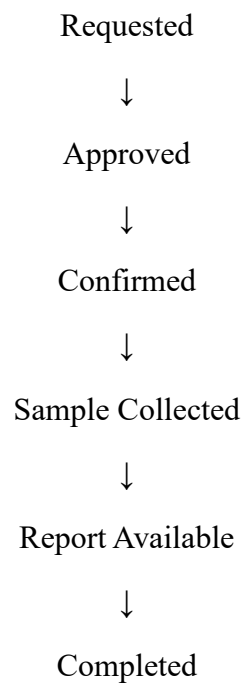
Appointment cancelled.

Actions:

- Update appointment status
- Process refund logic
- Update payment status
- Send cancellation confirmation

15. Appointment Status Lifecycle

The appointment follows a defined lifecycle:



An appointment may also move to:

Cancelled

depending on the cancellation process.

This lifecycle allows users and administrators to track the current state of every appointment.

The screenshot shows the ServiceNow 'APPOINTMENT TABLE' form. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', and a search bar. The form title is 'APPOINTMENT TABLE - Created 2026-09-02 21:44:15'. The form is divided into several sections: 'Appointment Number' (APPT-2026-51345), 'Status' (pending), 'Approval Date' (empty), 'Approval Status' (requested), 'Email' (empty), 'Total Amount' (empty), 'Payment Status' (Paid), 'Rejection Reason' (empty), 'Refund Amount' (\$ 0.00), 'Confirmation Sent' (empty), 'Test' (dfc3a6dac30bc790bc645fd1b4), 'Slot Time' (empty), 'Created By' (System Administrator), 'Scheduled Date' (2026-09-07 17:00:00), 'Patient' (f7ccee1ec34bc790bc645fd1b4), 'Notes' (empty), 'Approved By' (empty), 'Test Name' (empty), 'Test Code' (empty), 'Category' (-- None --), and 'Description' (empty). The 'Status' dropdown menu is open, showing options: -- None --, Cancelled, Completed, Confirmed, InProgress, Scheduled, and pending (selected). The 'Test' field has a search icon and a refresh icon. The 'Patient' field has a search icon and a refresh icon. The 'Approved By' field has a search icon and a refresh icon.

16. Notifications

Automated email notifications are configured for important events.

Notifications include:

- Appointment booking confirmation
- Payment confirmation
- Approval notification
- Rejection notification
- Appointment reminder
- Report availability
- Cancellation confirmation
- Refund confirmation

Dynamic information can be included in emails, such as:

- Patient name
- Test name
- Appointment number
- Appointment date
- Appointment time
- Payment amount
- Report information

17. Scheduled Jobs

Scheduled jobs are used for time-based operations.

Examples

Appointment Reminder

Automatically identifies upcoming appointments and sends reminder notifications.

Missed Appointment Update

Identifies appointments that have passed without completion and updates their status.

Expired Booking Cleanup

Identifies expired pending bookings and releases the associated slots.

Scheduled jobs improve automation and reduce manual administrative work.

18. Security & ACL Configuration

Since the system handles sensitive medical information, security is a major component.

Access Control Lists are configured at table and field levels.

Patient Access

Patients can:

- View their own appointments
- View their own reports
- Create appointments

Patients cannot:

- View another patient's information
- Modify administrative records
- Access another patient's report

Lab Technician Access

Lab Technicians can:

- View assigned appointments
- Update sample collection status

- Upload reports

Administrator Access

Administrators can:

- Manage patients
- Manage appointments
- Manage diagnostic tests
- Manage reports
- Configure workflows
- Manage security

19. Security Testing

Security is validated using:

Impersonation

The administrator can impersonate different users to verify their access permissions.

Debug Security Rules

ServiceNow security debugging can be used to identify why a user is allowed or denied access.

Role-Based Testing

Separate test users are created for:

- Patient
- Receptionist
- Lab Technician
- Administrator

This ensures that each role receives only the required permissions.

20. Error Handling

The application handles multiple edge cases.

Scenario 1 – Occupied Slot

If a patient attempts to book an already occupied slot:

Result: Booking is rejected.

Scenario 2 – Missing Information

If mandatory fields are empty:

Result: Validation message is displayed.

Scenario 3 – Duplicate Booking

If the same patient attempts to create a duplicate booking:

Result: System prevents the duplicate record.

Scenario 4 – Unauthorized Report Access

If a patient attempts to access another patient's report:

Result: ACL prevents access.

Scenario 5 – Appointment Cancellation

When an appointment is cancelled:

- Appointment status is updated.
- Payment/refund status is processed.
- Notification is sent.

21. Testing & Validation

The application is tested using multiple test scenarios.

Test Case	Expected Result
Select diagnostic test	Test selected successfully
Select available slot	Slot selected
Select occupied slot	Booking prevented
Submit valid appointment	Appointment created
Missing mandatory field	Validation error
Approve appointment	Status updated
Reject appointment	Status updated
Upload report	Report stored

Test Case	Expected Result
Download report	Authorized patient can access
Unauthorized report access	Access denied
Cancel appointment	Status changed to Cancelled
Notification trigger	Email generated

22. Reports & Dashboards

Reports and dashboards can be configured for administrators.

Possible reports include:

- Total appointments
- Completed appointments
- Cancelled appointments
- Pending appointments
- Test-wise bookings
- Revenue/payment summary
- Report completion status

Dashboards provide administrators with a centralized view of system activity.

23. Deployment

After development and testing, configurations are moved between ServiceNow instances using **Update Sets**.

Deployment Process

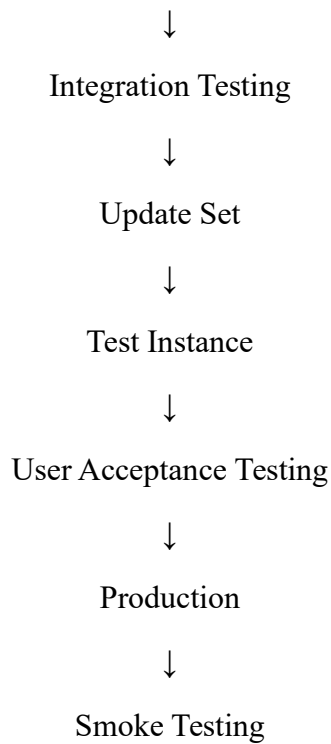
Development



Configuration



Unit Testing



24. Final Outcome

The Medical Diagnostic Test Center Portal successfully provides an automated platform for managing diagnostic center operations.

The system:

- Reduces manual appointment processing
- Prevents duplicate bookings
- Automates workflows
- Provides real-time appointment tracking
- Enables secure report management
- Automates notifications
- Implements role-based security
- Improves administrative efficiency

25. Future Enhancements

- Online Payment Gateway
- SMS Notifications
- Mobile Application
- Hospital Integration

- Advanced Analytics
- AI-Assisted Diagnosis

26. Conclusion

The **Medical Diagnostic Test Center Portal** demonstrates how ServiceNow can be used to automate a complete healthcare diagnostic workflow.

By combining Service Portal, Custom Tables, Client Scripts, GlideRecord, Flow Designer, Notifications, Scheduled Jobs, ACLs, Reports, and Dashboards, the system provides a centralized solution for appointment booking, processing, report management, and status tracking. The implementation improves operational efficiency while maintaining controlled access to sensitive information.

Overall, the project provides a secure, automated, scalable, and user-friendly platform for managing diagnostic test center operations.